

EirVah: An Open-Source Hybrid-Cloud Framework for Industrial Cyber-Physical Systems

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Abstract. The adoption of Industry 4.0 technologies requires Cyber-Physical System frameworks that support scalable data exchange, hybrid cloud deployment, and compliance with industrial standards. However, many CPS frameworks in industry remain largely conceptual, vendor- or environment-dependent, insufficiently scalable, and operationally complex, limiting practical adoption across heterogeneous infrastructure environments and increasing long-term operational costs. This paper presents EirVah, a fully open-source hybrid-cloud CPS framework designed for simplicity, deployment portability, and cost-efficient operation. EirVah integrates the ISA-95 Unified Namespace (UNS) as a core architectural construct and leverages Kubernetes-based orchestration and infrastructure-as-code (IaC) to enable consistent deployment across private and public cloud environments while minimizing vendor lock-in. Through model-based architectural analysis and structured hybrid deployment scenarios, the framework demonstrates scalable event-driven communication, deployment reproducibility, operational transparency, and reduced vendor dependency, contributing to a lower total cost of ownership across hybrid infrastructures.

Keywords: Cyber-Physical Systems, Industry 4.0, ISA-95, Unified Namespace, Hybrid Cloud, Open Source, Kubernetes, Infrastructure as Code, Artificial Intelligence

1 Introduction

Distributed systems have long been central to software engineering, evolving from monolithic applications to client-server models, service-oriented architectures, and ultimately microservices, which enhance scalability and resilience through independently deployable services [1]. Advances in cloud computing, virtualization, containerization, and orchestration further accelerated this shift

by enabling elastic resource provisioning, deployment portability, and automated coordination across heterogeneous infrastructures. Cloud-native computing [2] encapsulates these principles, promoting applications composed of lightweight containerized microservices managed through automated orchestration.

While these developments improve modularity and scalability, they do not eliminate complexity—they transform it. Decomposition introduces challenges such as network uncertainty, partial failures, data consistency trade-offs, and operational overhead. The fallacies of distributed computing [3] and documented architectural anti-patterns [4] illustrate the recurring risks of unmanaged distributed complexity.

CPS are described through diverse definitions in the literature [5], but are broadly understood as engineered systems integrating physical and computational components within unified operational frameworks. By their nature, CPS are inherently distributed, making explicit management of distributed complexity a fundamental design concern. As a key enabler of Industry 4.0 [6], CPS integrate enterprise information systems with operational technology and device-level telemetry. Modern CPS must therefore support scalable telemetry ingestion, reliable distributed processing, and cross-organizational data exchange while addressing the structural challenges of distributed architectures.

In practice, CPS deployments frequently encounter vendor-specific infrastructure dependence, hybrid-cloud requirements, operational complexity, configuration drift, and escalating costs. Organizations in regulated sectors increasingly pursue CPS and Industry 4.0 initiatives, yet data sovereignty requirements, compliance obligations, and security considerations frequently restrict adoption of public cloud platforms. This leads to hybrid cloud deployment strategies in which critical components remain on-premise [7]. Nevertheless, many contemporary CPS frameworks lack infrastructure-agnostic deployment models, limiting their adaptability across heterogeneous cloud environments.

The ISA-95 Unified Namespace (UNS) pattern is gaining widespread adoption in industry, providing a standardized hierarchical structure for organizing industrial data. Yet in many implementations, UNS functions primarily as a messaging convention rather than as a fully integrated architectural foundation.

Contributions. This paper makes the following contributions:

1. **Architectural Gap Analysis.** We identify structural limitations in existing CPS frameworks regarding hybrid-cloud portability, vendor dependence, operational complexity, and cost, highlighting the absence of deployable open-source reference architectures integrating Unified Namespace principles with cloud-native orchestration.
2. **Open-Source Hybrid-Cloud Framework Design.** We present EirVah, an open-source CPS framework enabling consistent deployment across private and public cloud environments, emphasizing reproducibility and vendor neutrality through open technologies and Infrastructure-as-Code (IaC).
3. **UNS as a Core Architectural Construct.** We position UNS as a core architectural construct governing data structure, service interaction, and semantic consistency, rather than merely a messaging convention.

4. **Scenario-Based Architectural Validation.** We validate the framework across hybrid deployment scenarios, demonstrating portability, elastic processing, operational transparency, and resilience via Kubernetes-based orchestration.

2 Background and Related Work

CPS emerged to formalize the integration of computational and physical processes, with Industry 4.0 further accelerating their development and positioning Industrial CPS (ICPS) as a core pillar of the Fourth Industrial Revolution [8]. Research has predominantly focused on security and safety concerns [9], while numerous CPS frameworks have been proposed addressing conceptual, logical, architectural, and model-based perspectives [10]. Given its emphasis on hybrid-cloud portability, simplicity, reproducibility, and vendor neutrality, this work is situated within the architectural domain of CPS framework design. Although no standardized CPS framework currently exists [11], several architectural approaches have been proposed, including cloud-based, service-oriented, and multi-tiered models. Recent studies highlight ongoing gaps in integrating distributed ICPS with virtualization technologies for industrial applications [12].

ISA-95 defines the integration between enterprise systems and manufacturing control systems, providing a structured framework for aligning operational technology (OT) and information technology (IT). It establishes a hierarchical model that separates shop-floor control from enterprise-level applications, traditionally restricting communication to adjacent layers (e.g., ERP systems do not directly interface with PLCs). The UNS is an architectural pattern that consolidates entities, resources, and devices within a centralized data model, enabling structured and decoupled data exchange [13].

Containers encapsulate applications and dependencies into portable, lightweight units, making them suitable for resource-constrained CPS environments [14]. Kubernetes orchestrates containerized workloads by automating deployment, scaling, networking, and lifecycle management to ensure reliable and elastic operation. Widely adopted in cloud environments [15], it also supports CPS requirements such as elasticity and fault recovery [16]. Recent work highlights Kubernetes as an effective platform for AI-intensive and distributed CPS workloads [17], while Cabral et al. [18] integrate it with the Arrowhead framework to enhance scalability and service orchestration. However, the architectural complexity of Arrowhead may introduce additional technical and operational overhead for industrial adopters.

Despite advances in CPS frameworks, Unified Namespace (UNS) architectures, and Kubernetes-based orchestration, important gaps remain. Existing CPS approaches insufficiently address hybrid-cloud portability and vendor-neutral deployment in industrial contexts. UNS research largely focuses on interoperability and data structuring, with limited attention to cloud-native deployment models. Moreover, the role of container orchestration within CPS architectures remains underexplored. As a result, the literature offers limited guidance on in-

tegrating CPS principles, UNS-driven design, and container orchestration within a streamlined and portable hybrid-cloud framework.

3 Research Methodology

This work adopts a design science research approach, resulting in the development of the EirVah architectural framework. Design science focuses on the systematic creation and evaluation of artifacts to address identified real-world problems through principled design and evaluation. The methodology comprises four stages:

1. **Problem Identification.** A structured review of CPS frameworks, Unified Namespace (UNS) architectures, and Kubernetes-based orchestration was conducted to identify limitations in hybrid-cloud portability, vendor neutrality, operational complexity, and cost.
2. **Architectural Abstraction.** CPS are abstracted as layered functional components to analyze integration across physical processes, messaging infrastructure, orchestration, and enterprise systems.
3. **Framework Design.** A hybrid-cloud, open-source reference architecture was developed emphasizing simplicity, Infrastructure-as-Code (IaC) reproducibility, and vendor neutrality.
4. **Hybrid Scenario Analysis.** Representative deployment scenarios were defined to assess feasibility, portability, and operational implications across heterogeneous environments.

4 CPS Abstraction and Architectural Principles

The literature presents numerous definitions of CPS [5]. A commonly adopted abstraction conceptualizes CPS as comprising three interrelated layers: physical, data, and decision [19]. This layered perspective provides a structured foundation for architectural analysis of CPS. The following diagram illustrates CPS abstracted:

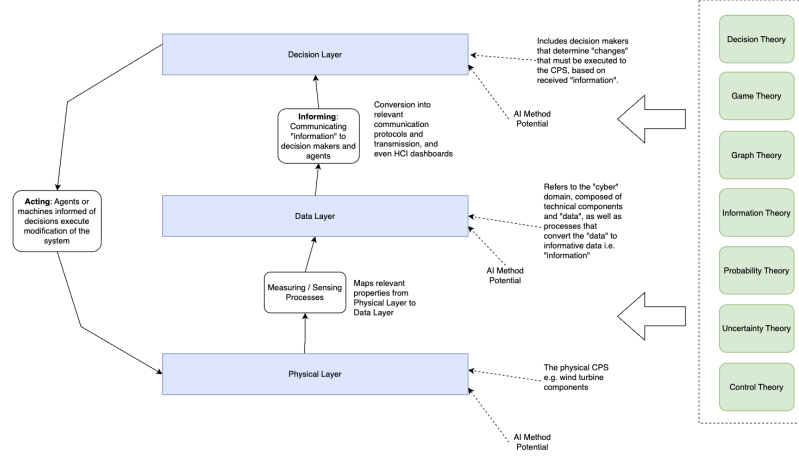


Fig. 1. Abstract Representation of CPS

The main points of the diagram are as follows:

1. **Physical Layer.** The physical layer comprises tangible system components such as industrial machinery, embedded sensors, and actuators. It generates quantitative and qualitative operational data through continuous or event-driven measurements via sensing and measurement mechanisms. In some cases, edge-based artificial intelligence supports low-latency analysis. The layer executes control commands received from the decision layer via actuators to effect corresponding physical changes in the system.
2. **Data Layer.** The data layer manages storage, processing, and transformation of raw inputs into structured information. It includes databases, integration services, and enterprise systems. Processed data is contextualized and selectively forwarded from the physical layer to the decision layer. Artificial intelligence techniques may support cleansing, structuring, classification, and pattern detection.
3. **Decision Layer.** The decision layer evaluates processed information from the data layer to generate control actions based on rules, optimization methods, analytical models, or AI techniques. Decisions are transmitted to the physical layer for execution. A human-in-the-loop mechanism may supervise or override automated decisions, enabling collaborative intelligence in safety-critical or uncertain contexts.
4. **Applied Theory** From an abstract perspective, Cyber-Physical Systems (CPS) are underpinned by multiple theoretical domains that collectively support modeling, decision-making, communication, and control in complex, interconnected environments. Decision, Game, Graph, and Information Theory, together with Probability, Uncertainty, and Control Theory, can all be applied to describe system structure, strategic interaction, data exchange, stochastic behavior, and feedback-driven dynamics within CPS.

5 EirVah Framework Architecture

The EirVah framework architecture is structured around the physical, data and decision layer enabling coordinated interaction through containerized applications orchestrated by Kubernetes across cloud and edge environments. The framework exclusively leverages open-source technologies to ensure cost efficiency while maintaining flexibility and scalability across both public and private cloud infrastructures. The following diagram presents the architectural overview.

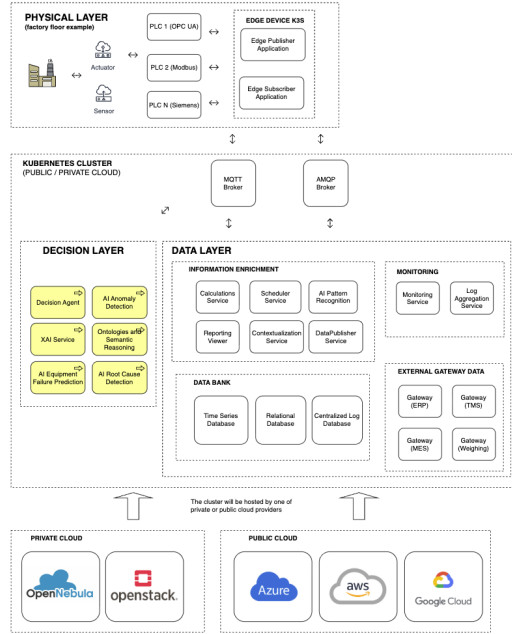


Fig. 2. EirVah Architecture Diagram

5.1 EirVah Physical Layer

The main component of the physical layer is an edge device responsible for protocol translation between industrial communication standards, and the framework's internal data broker protocols. The edge device has a set of microservices that implement an abstracted pipeline for protocol conversion, enabling the collection, transformation, and bidirectional exchange of data between the physical, data, and decision layers with UNS namespaces resolution during the transformation phase. Edge microservices are orchestrated using a lightweight Kubernetes

distribution (K3s). Physical changes within the CPS are affected through actuators controlled by PLCs, which are directed by applications running on the edge device.

5.2 EirVah Data Layer

The EirVah data layer comprises a collection of microservices and data storage components responsible for data management, integration, enrichment, and operational monitoring. It is structured into several functional areas. The Data Bank provides persistent storage through a time-series database for sensor telemetry, a relational database for event data and enriched information, and a centralized logging database that supports operational transparency and system observability. The External Gateway Data layer consists of integration interfaces that enable bidirectional communication with external enterprise subsystems such as ERP, MES, and SCADA platforms. Each integration is implemented through an abstract and extensible interface layer, allowing the framework to accommodate diverse third-party systems without architectural modification. The Information Enrichment layer is responsible for transforming raw data into structured and contextualized information. This includes services for time-series calculations, pattern recognition and clustering, data labeling, and contextual tagging in accordance with the Unified Namespace model. The Monitoring component continuously supervises the operational state of the framework by aggregating logs and performance metrics from all microservices into a centralized repository, enabling anomaly detection and alerting mechanisms.

5.3 EirVah Decision Layer

The EirVah decision layer receives contextualized information from the data layer and is responsible for generating control and optimization strategies based on that information. Decision agents within this layer apply a range of analytical techniques, from rule-based logic and statistical methods to advanced artificial intelligence models. These techniques support functions such as anomaly detection, semantic reasoning, predictive maintenance and equipment failure prediction, as well as root cause analysis.

In addition, the framework incorporates explainable artificial intelligence (XAI) mechanisms to provide transparency into model behavior and decision outcomes. This capability enables operators to understand the rationale behind automated decisions, thereby supporting trust, auditability, and regulatory compliance. Such transparency is particularly critical in highly regulated industrial domains, including pharmaceutical manufacturing, where traceability and validation of decision processes are mandatory.

5.4 EirVah Communication

Communication between the system layers is achieved through message brokers, enabling a publish-subscribe, asynchronous, event-driven architecture. This ap-

proach ensures loose coupling between layers and microservices, improving maintainability, scalability, and overall system flexibility. By decoupling producers and consumers, components can evolve independently while maintaining reliable communication.

MQTT (Message Queuing Telemetry Transport) MQTT is used to handle telemetry and time-series data usually generated by sensors, such as device temperature, battery level, and GPS location. Its lightweight design and efficient bandwidth usage make it well suited for constrained environments and large numbers of devices transmitting small, frequent messages.

AMQP (Advanced Message Queuing Protocol) AMQP is used for system-level events and inter-service communication. These messages represent discrete events or state changes, such as authentication events, workflow transitions, or fault notifications. AMQP provides reliable delivery and flexible routing, making it suitable for backend coordination. Decision layers also generate command messages that are sent through the messaging infrastructure to actuators, enabling controlled interaction with the physical CPS.

6 Hybrid Deployment Scenarios

This section evaluates EirVah portability across heterogeneous infrastructure environments. The analysis emphasizes deployment invariance: the platform architecture, service topology, and Infrastructure-as-Code (IaC) definitions remain unchanged, while only the infrastructure provisioning layer varies.

In private cloud environments, all components, Kubernetes clusters, message brokers, databases, and storage are deployed within a private infrastructure boundary. A managed Kubernetes cluster is provisioned on dedicated hardware, and EirVah is instantiated using cloud-agnostic IaC templates. Lightweight Kubernetes clusters may also be deployed at the edge using automated configuration, ensuring architectural consistency. This model prioritizes data sovereignty and operational control.

In public cloud environments, infrastructure resources are provisioned through a cloud provider, typically within a Virtual Private Cloud (VPC). Despite differences in underlying infrastructure, the same IaC templates and deployment strategy are used, preserving architectural invariance while reducing infrastructure management overhead.

In hybrid configurations, Kubernetes clusters are provisioned in both private and public environments. Services are distributed based on regulatory, latency, or cost considerations, but platform deployment remains identical across clusters. Edge integration follows the same automated strategy, maintaining structural uniformity.

Despite the distributed topology, the structural design of EirVah remains unchanged, demonstrating environmental independence and deployment portability. Across all scenarios, cloud-specific infrastructure provisioning is isolated

from platform deployment. This separation ensures that EirVah remains cloud-agnostic and portable without modification to its core architecture.

6.1 Open-Source Hybrid Deployment Strategy

Hybrid deployment across public and private environments is enabled through an open-source, cloud-agnostic strategy that preserves architectural portability and vendor neutrality. By standardizing Kubernetes configurations and isolating infrastructure-specific concerns, EirVah maintains structural consistency across heterogeneous environments.

- **Standardized Kubernetes Baseline.** All clusters share a consistent configuration, including a common Container Network Interface, DNS service, ingress controller, certificate management, and aligned network and security policies, ensuring predictable behavior across providers.
- **Cloud-Agnostic Deployment Model.** Workloads are deployed using Helm and Kustomize, maintaining a unified deployment definition while isolating environment-specific variations such as storage classes or load balancer settings.
- **Open-Source Platform Services.** Core services rely exclusively on open-source technologies (e.g., MQTT/AMQP brokers, PostgreSQL, Elasticsearch, Loki), preventing vendor lock-in and ensuring portability.
- **Cluster-Hosted Stateful Services.** Stateful workloads run as Kubernetes StatefulSets using portable storage abstractions rather than provider-managed services (e.g., Amazon RDS), preserving cloud vendor independence.
- **Provider-Neutral Observability and GitOps.** Monitoring leverages open-source tooling (e.g., Prometheus, Grafana), while infrastructure provisioning and application delivery follow Terraform-based provisioning and GitOps practices to ensure reproducibility.

This open source driven approach achieves hybrid portability by abstracting provider-specific infrastructure from the platform layer, enabling consistent multi-cloud deployment without modifying the core architecture.

7 Discussion

This work shows that hybrid-cloud portability in industrial CPS can be achieved by keeping the platform architecture consistent across environments while changing only the infrastructure provisioning layer. By combining Kubernetes orchestration with cloud-agnostic Infrastructure-as-Code and fully open-source components, EirVah enables deployment in both private and public clouds without architectural modification.

Portability depends on strict interface standardization. Even with managed Kubernetes services, differences in networking, storage, identity, and load balancing can create inconsistencies. EirVah mitigates this through a defined cluster

baseline covering networking, ingress, DNS, certificates, and observability, while isolating cloud-specific configurations through environment overlays. This reduces vendor lock-in and configuration drift but requires strong governance to prevent provider-specific features from entering application definitions.

The messaging-centric architecture strengthens loose coupling across CPS layers. MQTT handles time-series telemetry, while AMQP supports reliable backend coordination and event propagation. This separation enables independent scaling and asynchronous processing, improving flexibility. However, it increases governance demands, as topic hierarchies, schemas, and Unified Namespace semantics must remain consistent to avoid structural drift.

Relying on open-source technologies reduces vendor dependency and ensures consistent deployment across hybrid environments. However, operating stateful workloads such as databases and brokers within Kubernetes increases operational responsibility, particularly for backups, scaling, and upgrades. The economic benefit therefore lies in long-term control and flexibility rather than immediate cost savings.

In multi-cluster hybrid setups, stateless services are relatively portable, but state management and data locality remain key challenges. Cross-cluster communication introduces latency and trust boundaries, making it important to minimize synchronous dependencies.

Overall, EirVah demonstrates that hybrid-cloud CPS deployment is feasible through open-source standardization, declarative infrastructure, and a clear separation between provisioning and platform layers. The primary challenges are operational governance, state management, and semantic consistency within the Unified Namespace. AI-based methods can support anomaly detection, schema validation, and automated asset mapping, helping maintain semantic alignment as the system evolves.

8 Limitations and Future Work

Although EirVah provides an open-source and portable reference architecture, several limitations remain.

The current framework validation focuses primarily on architectural analysis and scenario-based reasoning rather than empirical benchmarking. Performance under high-throughput industrial workloads, particularly in latency-sensitive edge deployments, has not yet been systematically evaluated. Performance evaluation under realistic production conditions remains future work.

Running stateful services inside Kubernetes clusters improves portability, but it also increases operational complexity. Questions related to resilience, backup strategies, and storage performance need further investigation, especially in larger-scale production environments.

Hybrid multi-cluster orchestration challenges such as cross-cluster service discovery, networking, and federated identity management has not yet been validated and requires deeper study. Technologies such as service meshes or cluster

federation mechanisms may provide viable solutions, but their practical impact must be assessed.

AI components, including explainable AI mechanisms, are presently defined at a conceptual level rather than through fully implemented and evaluated pipelines. Their practical integration into industrial workflows, especially under regulatory and compliance constraints, remains an open area for further research and validation.

Future work will extend this foundation through empirical benchmarking in production-like environments, economic analysis of hybrid deployment models, validation of advanced orchestration strategies, and the deployment of operational AI pipelines that meet industrial regulatory and compliance requirements.

9 Conclusion

This paper presented EirVah, a fully open-source hybrid-cloud framework for CPS. The framework is grounded in an abstract, implementation-independent view of CPS that separates conceptual system modeling from deployment concerns. Building on this foundation, it addresses architectural gaps in existing CPS approaches by integrating ISA-95 Unified Namespace principles, Kubernetes-based orchestration, and Infrastructure-as-Code practices within a vendor-neutral deployment model built entirely on open-source technologies.

Through structured hybrid deployment scenarios, EirVah demonstrates that consistent architecture across public and private cloud environments can be achieved when infrastructure provisioning is abstracted from platform deployment. Its exclusive reliance on open-source components supports cost efficiency, portability, and long-term operational transparency.

By incorporating ISA-95 semantics into a cloud-native microservices architecture, EirVah provides a practical and reproducible reference model for industrial CPS environments seeking scalable and hybrid operation. This work lays the groundwork for further empirical evaluation and supports broader industrial adoption of portable CPS architectures aligned with Industry 4.0 principles.

Acknowledgements. Funded by the EU NextGenerationEU through the Recovery and Resilience Plan for Slovakia under the project No. 09I03-03-V04-00562.

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